



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201

ENGINEERING MATHEMATICS II

UNIT 3 & LP9 – VECTOR VALUED FUNCTION

1. VECTOR VALUED FUNCTION OF ONE VARIABLE:

A vector, each component of which is a function of same variable

$$F(t) = x(t)i + y(t)j + z(t)k$$

where $x(t)$, $y(t)$, $z(t)$ are component functions and t is a variable

Example: $F(t) = \cos t i + 2t^2 j + 3t k$

Derivative of $F(t) = -\sin t i + 4t j + 3k$

1.1 GEOMETRICAL INTERPRETATION MOTION ALONG A CURVE-VELOCITY, SPEED AND ACCELERATION

Vector algebra can be used to provide a geometric interpretation of velocity, speed, and acceleration in the context of motion along a curve.

VELOCITY:

Velocity is a vector quantity that describes the rate of change of position over time. It is defined as the derivative of the position vector with respect to time:

$$v(t) = \frac{d}{dt} r(t)$$

where $v(t)$ is the velocity vector at time t , $r(t)$ is the position vector at time t , and $\frac{d}{dt}$ is the derivative operator with respect to time.

Geometrically, the velocity vector represents the direction and magnitude of motion of the object at a particular point in time along the curve.

SPEED:

Speed is a scalar quantity that describes the magnitude of the velocity vector, or the rate at which the object is moving along the curve. It is defined as the magnitude of the velocity vector:

$$|v(t)| = \sqrt{v(t) \bullet v(t)}$$

where $|v(t)|$ is the speed of the object at time t , and $v(t) \bullet v(t)$ is the dot product of the velocity vector with itself.

ACCELERATION:

Acceleration is a vector quantity that describes the rate of change of velocity over time. It is defined as the derivative of the velocity vector with respect to time:

$$a(t) = \frac{d}{dt} v(t)$$

where $a(t)$ is the acceleration vector at time t , and $\frac{d}{dt}$ is the derivative operator with respect to time.

Geometrically, the acceleration vector represents the direction and magnitude of the change in velocity of the object at a particular point in time along the curve.

Problem 1: A particle moves along the curve given by $r(t) = 2t\vec{i} + 3t^2\vec{j}$ where $\vec{i}$ and $\vec{j}$ are the unit vectors in the x and y directions, respectively. Find the velocity and acceleration vectors at time $t = 2$.

Solution: The velocity vector can be found by taking the derivative of the position vector with respect to time:

$$v(t) = \frac{d}{dt} r(t) = 2\vec{i} + 6t\vec{j}$$

At $t = 2$, the velocity vector is:

$$v(2) = 2\vec{i} + 12\vec{j}$$

The acceleration vector can be found by taking the derivative of the velocity vector with respect to time:

$$a(t) = \frac{d}{dt} v(t) = 6\vec{j}$$

At $t = 2$, the acceleration vector is:

$$a(2) = 6\vec{j}$$

Problem 2. Find the velocity vector $(t) = 3t\vec{i} + 2t^2\vec{j} + \sin t\vec{k}$, Find the speed after $\frac{\pi}{4}$ seconds.

Solution:

The velocity vector can be found by taking the derivative of the position vector with respect to time:

$$v(t) = \frac{d}{dt} r(t) = 3\vec{i} + 4t\vec{j} + \cos \cos t \vec{k}$$

$$v\left(\frac{\pi}{4}\right) = 3\vec{i} + 4\left(\frac{\pi}{4}\right)\vec{j} + \cos \cos \left(\frac{\pi}{4}\right) \vec{k}$$

$$\text{Speed after } \frac{\pi}{4} \text{ seconds, is } |v\left(\frac{\pi}{4}\right)| = \sqrt{3^2 + \pi^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$$

$$= 4.4019.$$

Problem 3: A particle moves along the curve $r = (t^3 - 4t)\vec{i} + (t^2 + 4t)\vec{j} + (8t^2 - 3t^3)\vec{k}$, where t is the time. Find the speed and acceleration when $t = 2$.

Solution:

The velocity vector can be found by taking the derivative of the position vector with respect to time:

$$v(t) = \frac{d}{dt} r(t) = (3t^2 - 4)\vec{i} + (2t + 4)\vec{j} + (16t - 9t^2)\vec{k}$$

At $t = 2$, the velocity vector is:

$$v(2) = (3(2)^2 - 4)\vec{i} + (2(2) + 4)\vec{j} + (16(2) - 9(2)^2)\vec{k} = 8\vec{i} + 8\vec{j} - 20\vec{k}$$

Speed: The speed of the particle is the magnitude of its velocity vector:

$$|v(2)| = \sqrt{8^2 + 8^2 + (-20)^2} = \sqrt{768} = 27.7$$

The acceleration vector can be found by taking the derivative of the velocity vector with respect to time:

$$a(t) = \frac{d}{dt} v(t) = 6t \vec{i} + 2 \vec{j} + (16 - 18t) \vec{k}$$

At $t = 2$, the acceleration vector is:

$$a(2) = 6(2) \vec{i} + 2 \vec{j} + (16 - 18(2)) \vec{k} = 12\vec{i} + 2\vec{j} - 20\vec{k}$$

Therefore, at $t = 2$, the speed of the particle is approximately 27.7 units, and its acceleration vector is $12\vec{i} + 2\vec{j} - 20\vec{k}$

Problem 4: A rigid body is traveling along the x-axis in a rectangular coordinate system. Assume that the instantaneous position of the body may be represented by a function of $x(t) = 11t^2 - 2t^3$ in meter, in which t is time in second. We further Assume the mass of the body is negligible. Determine the vector functions of the velocity and acceleration of the moving rigid body.

Solution: We may determine the magnitude of the velocity vector function $v(t)$ and the magnitude of the acceleration vector function $a(t)$ as shown below

$$v(t) = \frac{d}{dt} x(t) = 22t - 6t^2$$

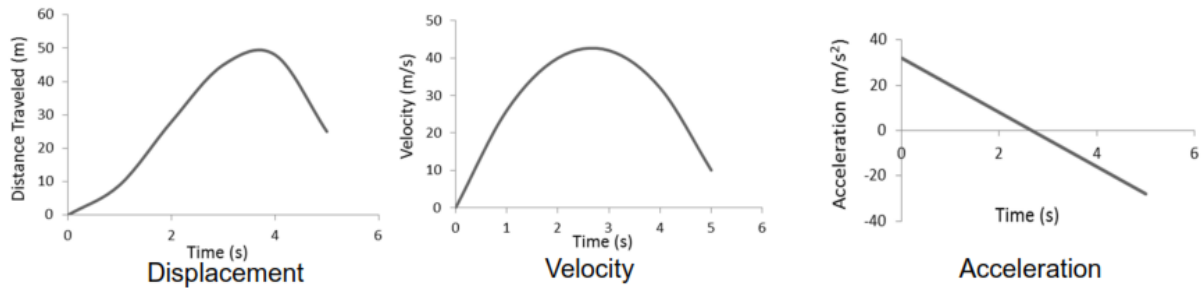
$$a(t) = \frac{d}{dt} v(t) = 22 - 12t$$

The vector functions of the velocity and acceleration of the moving rigid body thus can be expressed as

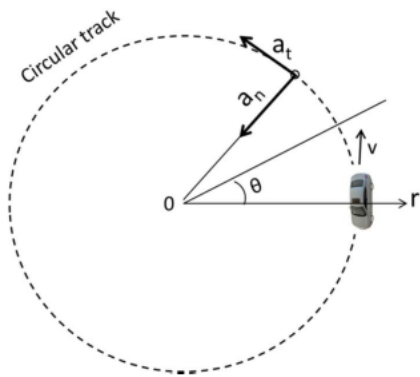
$$v(t) = (22t - 6t^2) \vec{i}$$

$$a(t) = (22 - 12t) \vec{i}$$

Graphical solution:



Problem 5: An automobile travels on a circular track with a radius of 25 m as shown in figure



If the magnitude of the velocity vector function of the vehicle is $v(t) = 1 + t^2$ and the starting location of the vehicle is P in the figure. Determine the magnitude of the acceleration of the vehicle at 4 seconds from standstill location.

Solution:

We are given the velocity vector function to be: $v(t) = 1 + 2t^2$ m/s, from which we may obtain the magnitude of the tangential component of the acceleration vector function to be

$$a_t = \frac{d}{dt} v(t) = 4t$$

$$a_n = -\frac{v^2}{r}, r \text{ is radius}$$

$$= - (1 + 4t^4 + 4t^2) / 25$$

The magnitude of the acceleration vector of the vehicle at $t=4s$ is

$$a(t) = \sqrt{a_t^2 + a_n^2} \text{ at } t = 4$$

$$= \sqrt{16^2 + (-43.56)^2}$$

$$= 46 \text{ m/s}^2$$

Problem 6: A particle moves along the curve given by the equation $r(t) = 4 \cos(t) \vec{i} + 4 \sin(t) \vec{j}$, where $\vec{i}$ and $\vec{j}$ are the unit vectors in the x and y directions, respectively. Find the velocity and acceleration vectors at time $t = \pi/4$.

To find the velocity vector, we need to take the derivative of the position vector with respect to time:

$$r'(t) = -4 \sin(t) \vec{i} + 4 \cos(t) \vec{j}$$

At time $t = \pi/4$, we have:

$$r'(\pi/4) = -2\sqrt{2} \vec{i} + 2\sqrt{2} \vec{j}$$

So the velocity vector at time $t = \pi/4$ is:

$$v(\pi/4) = r'(\pi/4) = -2\sqrt{2} \vec{i} + 2\sqrt{2} \vec{j}$$

To find the acceleration vector, we need to take the derivative of the velocity vector with respect to time:

$$a(t) = -4 \cos(t) \vec{i} - 4 \sin(t) \vec{j}$$

At time $t = \pi/4$, we have:

$$a(\pi/4) = -2\sqrt{2} \vec{i} - 2\sqrt{2} \vec{j} \text{ So the acceleration vector at time } t = \pi/4 \text{ is:}$$

$$a(\pi/4) = -2\sqrt{2} \vec{i} - 2\sqrt{2} \vec{j}$$

Problem 7: A rocket is traveling in a straight line in three-dimensional space, with position vector given by $\mathbf{r}(t) = (4t \vec{i} + 2t^2 \vec{j} - 3t^3 \vec{k})$ meters, where t is time in seconds. Find the velocity, speed, and acceleration of the rocket at time $t = 2$ seconds.

What is the position vector of the rocket?

What is the velocity vector of the rocket?

What is the speed of the rocket?

What is the acceleration vector of the rocket?

What is the magnitude of the acceleration vector?

Solution:

The position vector of the rocket is $\mathbf{r}(t) = (4t \vec{i} + 2t^2 \vec{j} - 3t^3 \vec{k})$ meters.

To find the velocity vector of the rocket, we differentiate the position vector with respect to time:

$$\mathbf{v}(t) = (4 \vec{i} + 4t \vec{j} - 9t^2 \vec{k}) \text{ m/s}$$

Therefore, the velocity vector of the rocket at $t = 2$ seconds is:

$$\mathbf{v}(2) = 4 \vec{i} + 8 \vec{j} - 36 \vec{k} \text{ m/s}$$

To find the speed of the rocket, we need to compute the magnitude of the velocity vector:

$$|\mathbf{v}| = \sqrt{4^2 + 8^2 + (-36)^2} = 37.4 \text{ m/s}$$

To find the acceleration vector of the rocket, we differentiate the velocity vector with respect to time:

$$\mathbf{a}(t) = d\mathbf{v}(t)/dt = (4 \vec{j} - 18t \vec{k}) \text{ m/s}^2$$

Therefore, the acceleration vector of the rocket at $t = 2$ seconds is:

$$\mathbf{a}(2) = 4 \mathbf{j} - 36 \mathbf{k} \text{ m/s}^2$$

To find the magnitude of the acceleration vector, we compute its magnitude:

$$|\mathbf{a}| = \sqrt{0^2 + 4^2 + (-36)^2} = 36.8 \text{ m/s}^2$$

Practice Problems:

1. Find the velocity vector, speed, acceleration if the position vector is $\mathbf{r}(t) = 3t\mathbf{i} + 2t^2\mathbf{j} + \sin(t)\mathbf{k}$
2. Find the velocity and acceleration of the position function $\mathbf{r}(t) = (2t-2)\mathbf{i} + (t^2+t+1)\mathbf{j}$ when $t = -1$.
3. An object moves in a helix with position function $\mathbf{r}(t) = (\cos t, \sin t, t)$ where distances are measured in meters and time is in minutes. Describe the object's speed and acceleration at time t .

Discussion questions

1. How can vectors be used to represent geometric shapes and transformations?
2. How can vectors be used to represent forces and motion in physics?
3. How do vectors relate to computer graphics and game development?